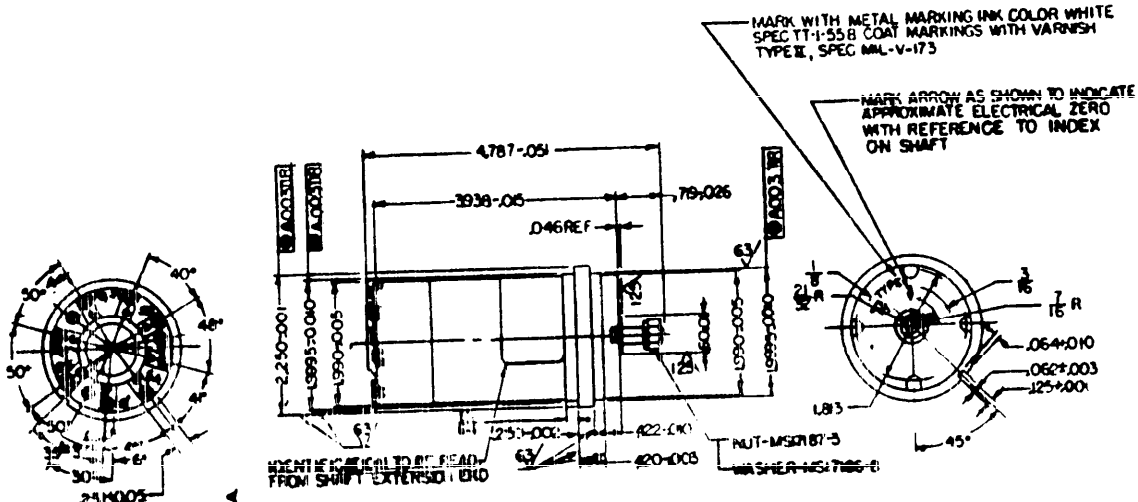
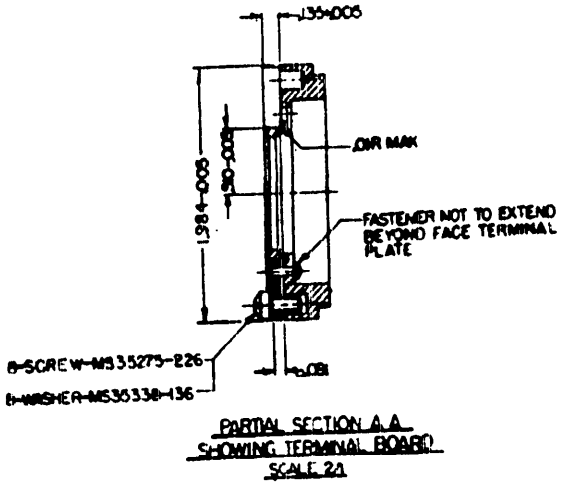
MILITARY SPECIFICATION SHEET

# RESOLVER, ELECTRICAL: TYPE 23R3N4A

The complete requirements for procuring the resolver described herein shall consist of this document and the issue in effect of Specification MIL-R-23417.




NOTE: THIS SPECIFICATION IS TO BE USED TO SUPPORT IN-SERVICE MATERIEL ONLY AND NOT TO BE USED FOR NEW DESIGN.




NOTES:
1-ECCENTRICITY OF SHAFT DIAMETER (SMOOTH PORTION) SHALL NOT EXCEED .001 TIR WHEN SHAFT IS ROTATED IN BEARINGS.
2-DIMENSIONS ARE IN INCHES. TOLERANCES WHEN NOT SPECIFIED ARE ±.005 ON DECIMALS, ±1/64 ON FRACTIONS AND ±1° ON ANGLES.
3-THE DESIGN OF THIS UNIT SHALL PROVIDE FOR THE TERMINAL BLOCK TO REMAIN FIXED WHEN TERMINAL SCREWS AND TERMINAL LUGS ARE REMOVED.



FSC 5990

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TABLE I

RESOLVER, ELECTRICAL: TYPE 23R3N4A

PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

| | Requirements | Value | Unit | Tolerance |
|---|---|---|---|---|
| 1. | Excitation winding | Stator | | |
| 2. | Rated frequency | 400 | Hz | ±1% |
| 3. | Input voltage | | volts (rms) | ±1% |
| | Maximum | 90 | | |
| | Test | 60 | | |
| | Minimum | 10 | | |
| 4. | Primary current | 260 | milliamps | nominal |
| 5. | Primary power | 1.9 | watts | nominal |
| 6. | D. C. resistance | | ohms | ±15% |
| | Rotor | 16.2 | | |
| | Stator | 4.84 | ohms | R±15%+jx10% |
| 7. | Impedance | | | |
| | $Z_{RO}$ | 45+j260 | | |
| | $Z_{SO}$ | 29+j230 | | nominal |
| | $Z_{RS}$ | 25+j23 | | |
| 8. | Transformation ratio | | | ±1% |
| | Output/input | 1.03 | % | |
| 9. | T. R. variation | | | maximum |
| | Due to voltage | 0.05 | % spread | |
| 10. | Equality of transformation ratios | | | |
| | Output/input | 0.5 | degrees | ±0.5 |
| 11. | Phase shift | | | |
| | Output/input | 1.3 | degrees | |
| 12. | Phase shift variation | | | |
| | Due to voltage | ±0.5 | minutes | |
| 13. | Interaxis error | ± 5 | % | |
| 14. | Functional error | ±0.2 | mV(rms) output | maximum |
| 15. | Null voltage | | | |
| | Fundamental | 31.8 | | |
| | Total | 63.6 | minutes | |
| 16. | Shift of resolver zero | | | |
| | With voltage | ±3.0 | | |
| | With frequency | ±1.0 | | |
| 17. | Harmonic distortion | 0.1 | % | maximum |
| 18. | Temperature rise | 20 | °C | maximum |
| 19. | Friction torque | 0.2 | oz. - in. | maximum |
| | @ -55°C | 0.4 | | maximum |



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TABLE I (Continued)

| Requirements | Value | Unit | Tolerance |
|---|---|---|---|
| 20. Radial play | 0.0010 (for reversal of 1/2 lb load) | inches | maximum |
| 21. End play | 0.0050 (for reversal of 1 lb load) | inches | maximum |

Custodian:
Army - MU

Review activities:
Army - EL, AV

User activities:
Army - ME

Preparing activity:
Army - MU

(Project 5990-A014)



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